



Nitrogen ion irradiation of carbon thin films using a dense plasma focus: enhanced doping and structural modifications

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Abstract Activated carbons (ACs) are widely used in supercapacitors and batteries due to their high surface area, well-developed porosity, cost-effectiveness, and scalability, making them ideal for commercial energy storage applications. In this study, carbon derived from coconut shells was processed into a thin film on a nickel substrate, synthesized via doctor blade deposition, and subjected to nitrogen ion beam irradiation (6, 12, and 24 shots) using a 3.0 kJ dense plasma focus (PF) device, operated at 2.54 kJ to produce carbon nitride. The PF device generated a high-energy ion beam (~ 72.40 keV) at 1.5 mbar within nanoseconds, with a flux of 7.2×10^{27} ions $\text{m}^{-2} \text{s}^{-1}$ and a fluence of 6.4×10^{19} ions m^{-2} . The Lee model code estimated a nitrogen ion beam energy of 71.0 keV, closely matching the measured 72.40 keV, confirming the model's reliability. EDX analysis revealed significant spectral changes, with nitrogen doping reaching 7.93% after 24 shots. The optimum doping content per shot is highest at lower shot counts, reaching 1.18% per shot for the 6-shot sample, and gradually decreasing with additional shots. FTIR confirmed nitrogen-related functional groups ($\text{C}\equiv\text{N}$, $\text{S}-\text{C}\equiv\text{N}$, and $\text{N}=\text{C}=\text{S}$), indicating successful incorporation of nitrogen. FESEM showed pores, bulges, blunt edges, and adatoms after nitrogen ion irradiation, indicating morphological changes. XRD of the 24-shot sample revealed new peaks at 52° and 76° , corresponding to the (004) and (110) planes, associated with nitrogen incorporation and surface modification. The crystallite size increased from 6.27 nm to 11.16 nm after 24 shots of nitrogen ion irradiation, indicating enhanced crystallinity. An interlayer spacing of 0.340 nm ($3.40 \text{ \AA}$), approaching the graphitic value of $3.35 \text{ \AA}$, was observed after 24 shots, indicating the onset of graphitization. These findings confirm effective nitrogen doping and significant structural modifications.

1 Introduction

Activated carbons (ACs) are widely used as active materials in supercapacitors and batteries due to their exceptional properties, including high surface area and a well-developed porous structure. These characteristics play a crucial role in determining the power and energy storage capacity of such devices [1, 2]. Additionally, their cost-effectiveness and scalability make them highly suitable for commercial energy storage applications [3].

Plasma-based techniques for AC doping are essential for enhancing performance by enabling precise control over pore size and structure. Beyond structural modifications, surface functionalization is critical for further improving AC properties. Recent studies indicate that nitrogen doping can significantly enhance the specific capacitance and electrochemical performance of ACs, making it a promising approach for next-generation energy storage systems [4–6].

Hetero-element doping improves the intrinsic conductivity and pseudocapacitive performance of carbon electrodes, leading to increased gravimetric and volumetric capacitances [7, 8]. Nitrogen doping modifies the electron distribution within the carbon matrix, facilitating efficient ion diffusion [9]. In carbon materials, nitrogen doping occurs via substitutional incorporation of nitrogen atoms, which act as electron donors, enhancing n-type conductivity [10]. Furthermore, nitrogen doping enhances electrolyte ion migration by increasing wettability at the electrolyte–carbon interface, thereby improving the rate capability and cycling performance of carbon electrodes [11].

Fluorine doping further refines the electronic properties of carbon materials due to fluorine's high electronegativity (3.98). It influences electron donor–acceptor dynamics and adjusts the pore structure [12]. Fluorine atoms contribute to energy storage and

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conversion by stabilizing carbon structures and introducing surface defects [13]. Co-doping with nitrogen and fluorine increases the number of active sites, significantly enhancing pseudocapacitive performance [8].

In this study, we explore a method for co-doping nitrogen and fluorine by utilizing fluorine-containing organic compounds, such as polyvinylidene fluoride (PVDF), as a binder in the fabrication of thin films, combined with nitrogen ion doping using a plasma focus (PF) device. Nitrogen-doped carbon thin films exhibit enhanced conductivity, making them ideal for electrical devices [14]. Additionally, they demonstrate superior wear resistance compared to amorphous carbon due to the presence of strong N–C bonds, which strengthen adhesion to the substrate [15, 16]. Nitrogen-doped carbon materials also possess high hardness, attributed to the shorter bond length of C–N bonds (1.47 Å) compared to sp^3 C–C bonds (1.54 Å) [17].

The PF device is a well-established source of energetic ions [18], capable of producing ions with energies ranging from 25 keV to 8 MeV [19]. It employs self-generated magnetic fields to compress plasma to extremely high densities ($\sim 10^{25}$ – 10^{26} m $^{-3}$) and temperatures (~ 1 – 2 keV) [20]. Due to its ability to deliver high-energy ion pulses within nanoseconds, the PF device has been widely used for material processing, including ion implantation, thin film deposition, and doping of various materials [21]. Previous studies have demonstrated its effectiveness in nitrogen and carbon ion implantation [22–25], as well as in modifying phase transitions of thin films [24].

Nitrogen-doped carbon has emerged as a promising electrode material due to its high electrochemical stability, tunable bandgap, large surface area, and exceptional thermal and chemical resilience [26, 27]. Its affordability and environmental friendliness further position it as a strong candidate for next-generation lithium-ion batteries (LIBs) and supercapacitors [28].

However, conventional nitrogen doping methods, such as thermal treatment with nitrogen precursors, often suffer from low doping efficiency, uncontrolled incorporation, structural damage, and lengthy processing times. Similarly, continuous techniques, like plasma-enhanced chemical vapor deposition (PECVD) and radio frequency (RF) magnetron sputtering, exhibit a slow doping rate during deposition, as dopant atoms incorporate into the growing film. Additionally, these methods often suffer from poor adhesion compared to PF technique. Meanwhile, ion implantation requires significantly longer processing times to achieve a comparable doping level [24, 29].

Gold nanorods exhibit pronounced shifts in localized surface plasmon resonance (LSPR) with changes in aspect ratio [30], underscoring the sensitivity of plasmonic behavior to surface electronic structure and charge distribution. Although this phenomenon pertains to metal nanostructures, it conceptually supports the idea that plasmonic responses in carbon-based materials can also be modulated through surface electronic alterations. In this context, nitrogen doping via PF offers a promising route to energetically modify surface electron density—an effect that passive methods, such as PECVD or thermal treatment, typically fail to achieve. Although not directly focused on nitrogen-doped carbon, modeling studies of transition metal dichalcogenide (TMDC)–carbon nanostructures for fuel cells [31] emphasize the critical role of interfacial contact and electronic coupling in determining device performance. This principle is likewise relevant to carbon-based systems, where conventional nitrogen doping often lacks the surface activation necessary to promote strong interfacial bonding or controlled electronic tuning. In contrast, PF-based doping introduces energetic surface modification, enabling improved chemical bonding and reduced contact resistance. Supporting the broader efficacy of energetic plasma processes, Gholipour et al. [32] reported strong adhesion and uniform porosity in plasma-deposited GO/PANI/CuO films—key indicators of high-quality interfacial contact. While their work does not involve nitrogen doping, it reinforces the advantage of plasma-based techniques over conventional approaches such as PECVD, RF sputtering, and thermal treatment in producing well-integrated interfaces with desirable morphological and electrical properties.

Pulsed ion irradiation using a PF device offers a superior alternative, enabling rapid and efficient doping within an ultra-short timeframe (hundreds of nanoseconds). PF doping delivers high-energy ion irradiation without prolonged thermal exposure, allowing for more controlled and efficient nitrogen incorporation. Despite these advantages, research on PF-assisted nitrogen doping for doctor blade-deposited carbon films derived from coconut carbon shells remains unexplored.

This study addresses the existing gap by employing a PF device to achieve nitrogen doping in carbon thin films derived from coconut carbon shells. We further demonstrate that nitrogen ion irradiation induces atomic rearrangement, transforming amorphous carbon into a partially graphitized structure—an effect not previously reported in PF-assisted nitrogen doping of carbon materials. The structural, chemical, and morphological modifications resulting from nitrogen ion doping are thoroughly investigated, along with their impact on the electrochemical performance of the films for energy storage applications.

Watanabe et al. [33] studied nitrogen ion irradiation of bulk and thin film amorphous carbon using a bucket-type ion gun, reporting changes in surface roughness and composition. However, their ion gun operated at an energy of 0.2 keV, significantly lower than that of a PF device. Karamat et al. [24] successfully doped ZnO with nitrogen using a 2.94 kJ dense PF device, while Aghamir et al. [16] investigated the effect of deposition angle on amorphous carbon nitride thin films using PF irradiation, achieving a nitrogen doping level of 3.62%. In contrast, our study achieved a significantly higher nitrogen doping level of 7.93% in carbon thin films using a minimal number of PF shots.

Fig. 1 Schematic illustration of the electrode preparation process. Coconut shell-derived activated carbon, acetylene black, and PVDF binder are mixed in an 8:1:1 weight ratio and ground into a fine powder. Ethanol is added to form a homogeneous slurry, which is then coated onto a Ni foam substrate using the doctor blade method. After drying under controlled temperature and time conditions, a uniform electrode film is obtained for PF nitrogen ion irradiation

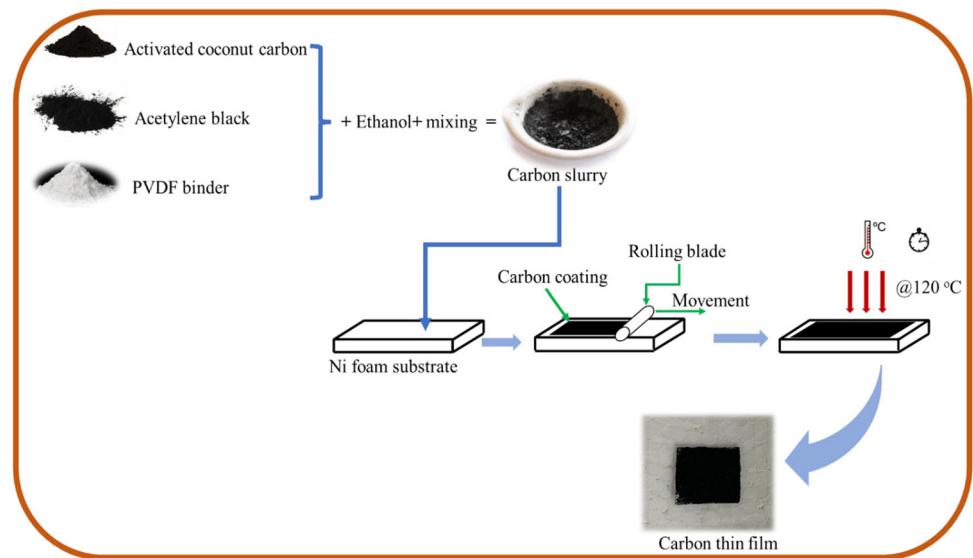


Table 1 Parameters utilized in the PF nitrogen ion doping

Parameter	Value
Capacitance	30 μF
Charging voltage	13 kV
Inductance	156 nH
Anode radius	0.95 cm
Cathode radius	3.2 cm
Anode length	16 cm
Resistance	21.4 m Ω

2 Experimental method

2.1 Synthesis and deposition of activated carbon

The AC used in the experiment was derived from coconut shells and processed into activated coconut carbon, as detailed in previous work by Wante et al. [34]. It was then crushed into a fine powder. The electrode film was prepared by applying a slurry mixture of coconut carbon shells, acetylene black, and PVDF in a weight ratio of 8:1:1, dissolved in ethanol. The components were manually ground using a mortar and pestle to achieve a fine powder. Ethanol was then added dropwise, and the mixture was thoroughly blended to form a homogeneous slurry. PVDF acted as both a binder and a fluorine dopant, promoting the formation of C-F bonds, which enhances the electrical conductivity and electrochemical performance of the carbon material, thereby increasing energy density [35].

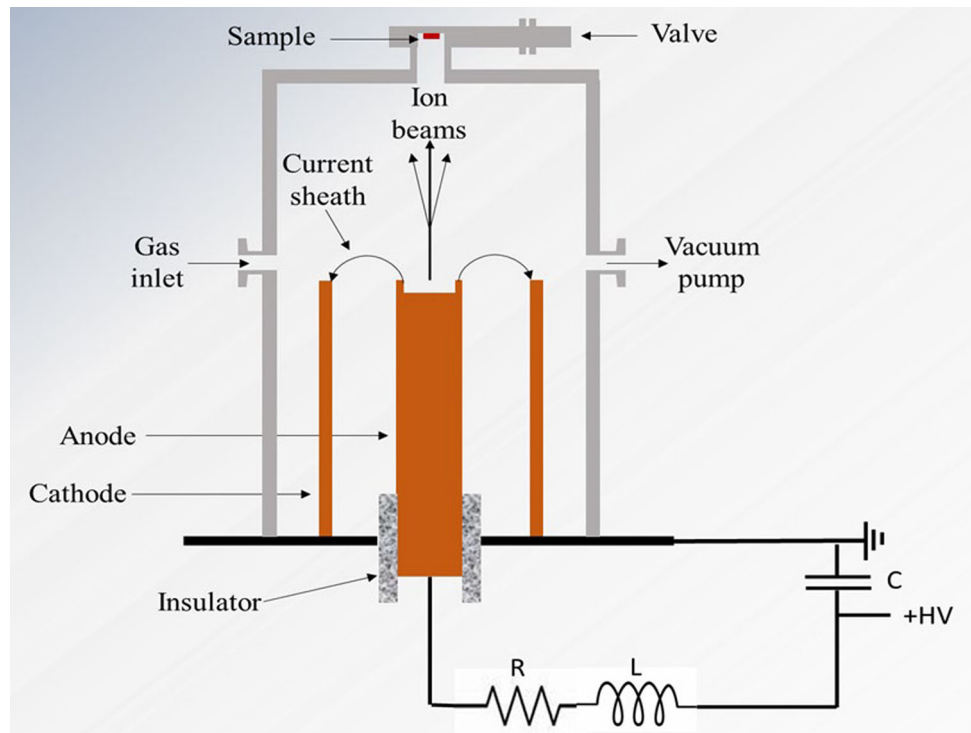
The slurry was then evenly applied to an 8 mm $\times$ 8 mm $\times$ 1 mm nickel foam sheet using the doctor blade technique to ensure a uniform coating. The samples were subsequently dried in an oven at 120 °C overnight to ensure complete ethanol evaporation, leaving behind a pure carbon coating on the nickel foam substrate. Figure 1 illustrates the doctor blade-fabricated thin film description.

2.2 PF device description for nitrogen ions irradiation

A 3.0 kJ UNU/ICTP PF device, operated at 2.54 kJ, is used as the high-energy nitrogen ion source for carbon thin film irradiation. Table 1 provides a detailed summary of the specific parameters and characteristics of the device. The UNU/ICTP PF device features six (6) symmetrically positioned cathodes around the anode. The central electrode is a flat-ended anode, 16 cm long and 1.9 cm in diameter, surrounded by six copper rods (16 cm long, 0.95 cm in diameter). The cathode electrodes are mounted on a copper baseplate and separated from the anode by a Pyrex insulator sleeve (2 cm thick, 5 cm long, 2.6 cm in diameter). The vacuum chamber is connected to a rotary pump, a vacuum gauge, and a nitrogen gas (99.999% purity) cylinder, all secured with vacuum valves and O-ring seals.

The Faraday cup detector is commonly used with the PF device to determine ion properties by accurately measuring ion current density, which enables the computation of their time of flight and energy [36]. Figure 2 shows the schematic configuration of the experimental setup for the UNU/ICTP PF device.

Fig. 2 Schematic setup for the UNU/ICTP PF device used for ion beam irradiation. The system comprises a coaxial electrode configuration with a hollow central anode and surrounding cathode, separated by an insulator



In a PF device, ion acceleration begins when capacitor banks discharge, applying a high voltage—typically in the range of tens of kilovolts (kV)—across the electrodes. This creates a strong electric field that induces electrical breakdown of the nitrogen gas (N_2) within the discharge chamber, ionizing the gas and generating free electrons and ions. Electron impact collisions further dissociate nitrogen molecules into atomic nitrogen ($N + N$) and ionize them to produce nitrogen ions such as N_2^+ , N^+ , and in some cases, doubly ionized species. The resulting current filaments are accelerated toward the anode by the Lorentz force [37], eventually converging along the central axis near the anode tip to form a high-density, high-temperature plasma pinch. At this stage, the plasma reaches extreme conditions, with temperatures on the order of $\sim 10^6$ K and particle densities ranging from 10^{18} to 10^{20} m^{-3} .

As the pinch becomes unstable and collapses, intense electric fields are generated, further accelerating nitrogen ions to high energies. Simultaneously, electrons bombard the anode surface, producing intense X-rays and causing localized evaporation at the anode tip. The resulting high-energy ions are emitted predominantly in the direction opposite to the electron flow and interact with both the target material and residual unionized nitrogen gas in the chamber. These interactions are crucial for surface modification, including nitrogen incorporation and the formation of carbon nitride films on substrates placed at the top of the anode. To reduce anode evaporation and contamination, a hollow anode is typically employed, allowing focused ion beam delivery to the carbon sample. This facilitates compound layer nucleation through sublimation and promotes uniform film growth.

In a typical PF device, ion beam energies range from tens of keV to MeV, while the device operates in the range of several kJ to MJ [21, 38]. The Faraday cup (FC) operates in the biased ion collector (BIC) mode with a biased voltage of -45 V. To prevent overheating of the doped films, the focusing action was repeated at 5-min intervals per shot. The carbon thin film samples were placed 38 cm from the anode tip, at the same position as the ion beam collector, and were exposed to nitrogen ion irradiation for 6, 12, and 24 shots.

A Yokogawa digital oscilloscope (DL7480) was used to capture the current, voltage, and ion signals. The interval between the plasma pinch formation (indicated by the X-ray signal) and the detection of ions (measured by the Faraday cup) defines the ion time of flight, as described by Eq. (1) [21];

$$t_{TOF} = t_{ion} - t_{SXR} \quad (1)$$

where t_{ion} is the peak time at the ion peak, and t_{SXR} is the time at the X-ray peak.

One can express the average energy of the nitrogen ions as follows [22];

$$E = \frac{1}{2}mv^2 = \frac{1}{2}m_N \left(\frac{x_{FC}}{t_{TOF}} \right)^2 \quad (2)$$

where m_N is the nitrogen mass and x_{FC} is the distance from the faraday cup to the anode tip.

Multiple experimental shots were performed to determine the optimal operating parameters for the UNU/ICTP PF device. The optimal configuration was identified as a 30- μ F capacitor charged to 13 kV with a nitrogen gas pressure of 1.5 mbar, determined

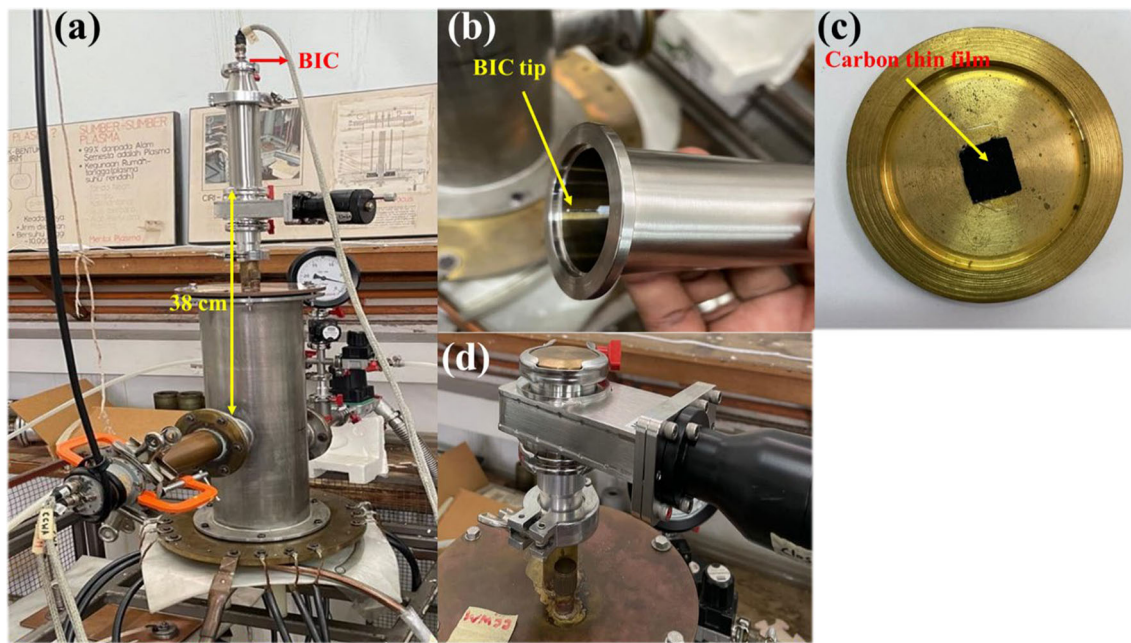


Fig. 3 **a** PF chamber showing the distance from BIC tip to the anode tip; **b** Showing BIC tip before installing; **c** Sample attached to NW 40 cover; **d** Installed cover with sample for ion irradiation

through repeated testing. The performance of the PF device is critically influenced by the working gas pressure, as it directly affects the generation of strong, well-focused plasma pinches and flux, indicating that a well-focused plasma pinch was achieved at 1.5 mbar. The vacuum chamber was initially evacuated to 5×10^{-3} mbar using a rotary pump to minimize residual gas contamination. High-purity nitrogen gas was then introduced to a higher pressure and gradually degassed to the optimal pressure. To achieve the ideal plasma pinch conditions, preliminary shots were conducted prior to initiating the doping process. Strong and stable pinch signals were observed after four (4) initial shots. Figure 3 depicts the pictorial view of the PF chamber used for ion irradiation.

To evaluate the influence of ion flux on the carbon thin film, it is essential to measure parameters such as ion flux and fluence per pulse.

The ion flux and fluence are given in Eqs. (3) and (4), respectively [39];

$$\text{Flux}, J_b(\text{ions m}^{-2}\text{s}^{-1}) = 2.75 \times 10^{15} \frac{f_e \ln(b/r_p) I_{\text{pinch}}^2}{M^{1/2} Z_{\text{eff}}^{1/2} r_p^2 U^{1/2}} \quad (3)$$

$$\text{Fluence}(\text{ions m}^{-2}) = 2.75 \times 10^{15} \cdot \tau \cdot \frac{f_e \ln(b/r_p) I_{\text{pinch}}^2}{M^{1/2} Z_{\text{eff}}^{1/2} r_p^2 U^{1/2}} \quad (4)$$

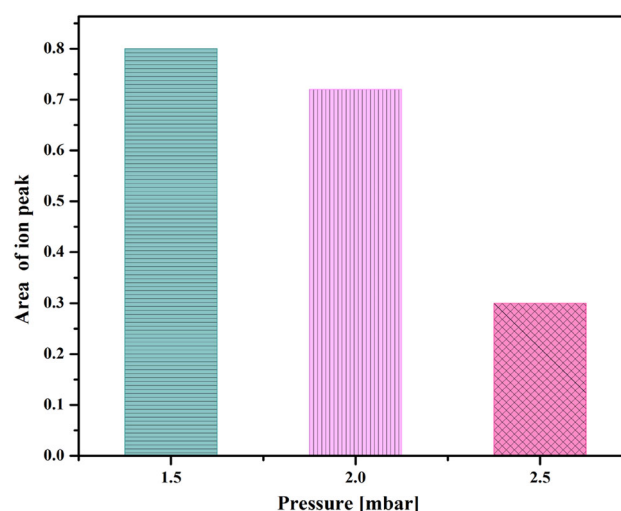
where Z_{eff} represents the effective charge, r_p is the pinch radius, M denotes the mass number of ions, τ denotes pinch duration s, I_{pinch} represents pinch current, and U denotes diode voltage.

The values of the ion flux and fluence for each situation are determined by fitting the current waveform and calculating the values using the Lee model code.

3 Film characterization

To explore the properties of the as-deposited and nitrogen-exposed thin films, a variety of analytical techniques were utilized, each designed to assess specific material characteristics. Scanning electron microscopy–energy-dispersive X-ray spectroscopy (SEM–EDX) analysis, using a TM3030 Hitachi microscope, was conducted to observe and quantify the elemental compositions across the thin films. Operating at 15 kV, the EDX system provided detailed elemental spectra and mapping analyses. Field emission scanning electron microscopy (FESEM (FEI QUANTA FEG 650 S)) offered high-resolution imaging to examine the surface morphology of both exposed and unexposed samples, revealing structural details and documenting any exposure-induced changes. Fourier transform infrared spectroscopy (FTIR), performed with a PerkinElmer 97,987 spectrometer, identified and characterized the functional groups, providing insights into the chemical bonds and molecular structures present in the samples. Finally, high-resolution

Fig. 4 Area under curve for the BIC at varying pressure with 13 kV charging voltage



X-ray diffractometer (HR-XRD) analysis, using an EMPY-REAN diffractometer with Cu-K α radiation at 5°/min, 5 kV, and 30 mA, determined the crystallographic structure of the samples. The diffraction patterns were used to calculate particle diameters (D) using Eq. (5) and assess the correlation between structural properties and sample performance [40, 41];

$$D = \frac{0.9\lambda}{\beta \cos \theta} \quad (5)$$

The diffraction angle is denoted by θ , the peak width is denoted by β , and the wavelength of the Cu-K α X-ray emission, which is 1.5 Å, is represented by λ .

The interlayer spacing was also calculated using the Bragg's law [42]:

$$d_{hkl} = \frac{\lambda}{2 \sin \theta} \quad (6)$$

where d is the interlayer spacing and hkl is the crystal plane.

4 Results and discussion

4.1 PF characteristics

After conducting multiple shots, we identified the optimal operating pressure in our PF device to be 1.5 mbar, which produced a strong ion beam signal as well as optimum area under the curve (Fig. 4). At this optimized pressure, we characterized the nitrogen ion beam, analyzing key parameters such as beam energy (E), flux, fluence, and the area under the curve. By employing Fig. 5b alongside Eqs. (1) and (2), E at this pressure was computed from the measured data to be ~ 72.40 keV.

In Fig. 5b, two peaks are observed: the first, corresponding to X-ray emission at time t_0 , aligns with the voltage peak, indicating intense electromagnetic interactions. The second peak, representing ion emission at time t_1 , occurs almost simultaneously with the X-ray emission and relates to the dynamics of the plasma sheath. The ion beam energy was accurately determined by measuring the flight time over a known distance (38 cm).

Figure 5a shows current and voltage waveforms at 1.5 mbar, providing insights into plasma characteristics, including electric and magnetic field interactions. These waveforms reveal the compression and radiative processes during plasma evolution [43], with the Lee model accurately fitting the computed and measured current signals (Fig. 5c), yielding an energy value of 71.0 keV. This energy value closely aligns with the measured energy of 72.40 keV, demonstrating the reliability of the model. Furthermore, Fig. 5c shows a strong agreement between the computed and measured current signals, further validating the model's accuracy. The model parameters ($f_m = 0.03$, $f_c = 0.7$, $f_{mr} = 0.18$, and $f_{cr} = 0.85$) were critical for achieving this fit [19]. Ion flux and fluence were computed to be 7.2×10^{27} ions $m^{-2} s^{-1}$ and 6.4×10^{19} ions m^{-2} , respectively, highlighting the high intensity and density of the ion beam.

4.2 Morphological analysis

The morphology of both as-deposited and nitrogen-irradiated carbon thin films was analyzed using FESEM, revealing structural modifications induced by nitrogen ion irradiation in a PF device. Figure 6a presents the as-deposited sample, characterized by

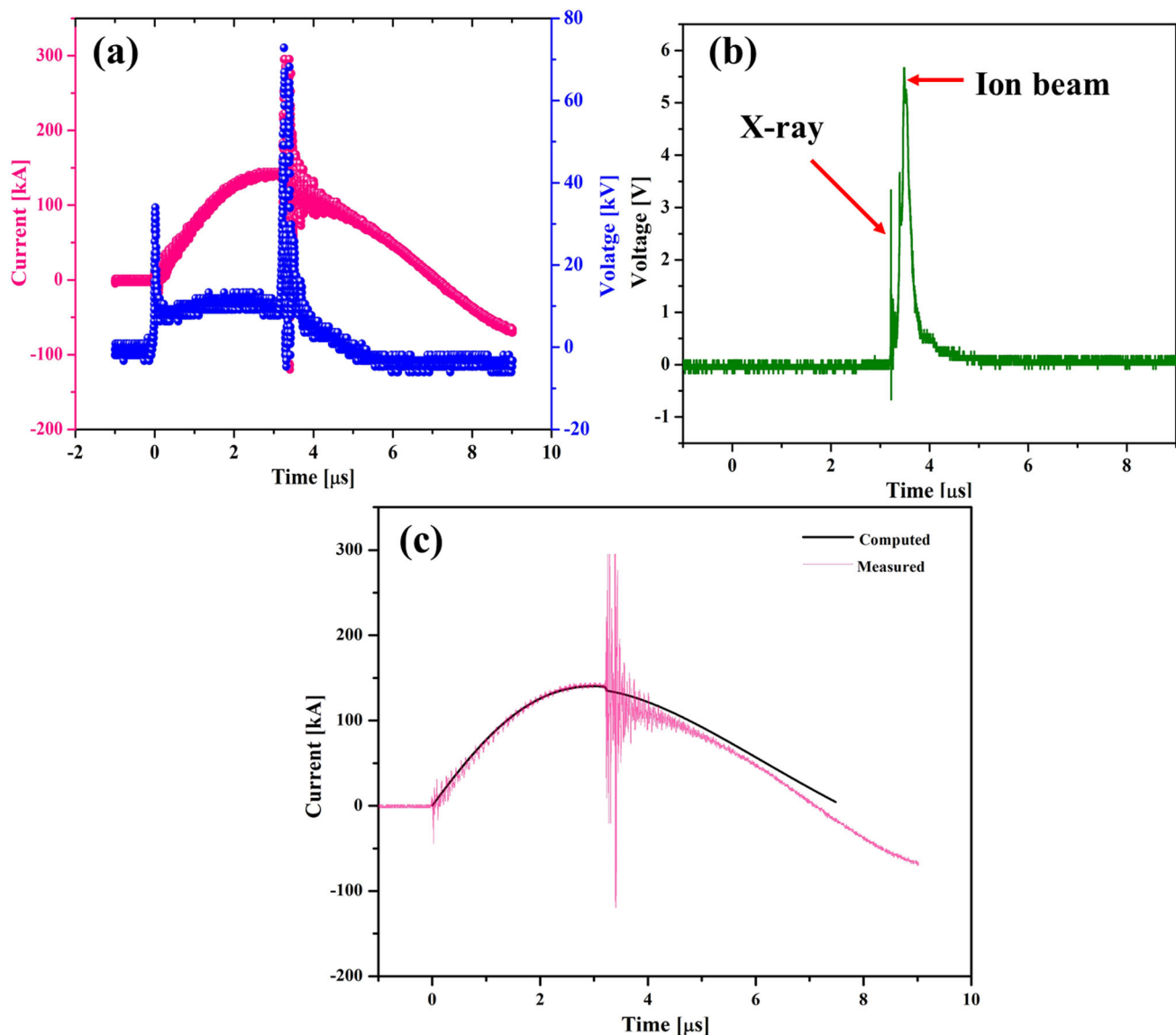


Fig. 5 **a** Current and voltage signals; **b** Ion beam signal at a pressure of 1.5 mbar recorded by BIC; **c** Fitted measured and computed current signal using Lee model code

sharp edges and a relatively rough morphology, indicative of an unmodified carbon structure. The absence of significant defects or distortions suggests that the initial film maintains its intrinsic microstructural properties.

Upon nitrogen ion irradiation (6 shots; Fig. 6b), distinct morphological transformations become evident. The emergence of nanoscale pores, semi-blunt edges, and surface bulges indicates significant ion–surface interactions. These features result from the interaction of energetic nitrogen ions with surface atoms, causing localized irradiation and material redeposition [44]. The formation of pores is particularly significant, as it enhances electrolyte penetration and increases the accessibility of active sites, which are crucial for improving electrochemical performance [45].

As the number of PF shots increases, the morphology undergoes further evolution. In Fig. 6c, corresponding to 12 shots, the distinct pores observed in Fig. 6b disappear, replaced by compacted atomic clusters. The increased ion flux enhances collision energy, promoting atomic mobility and facilitating the formation of a denser microstructure [46]. This transition suggests that prolonged nitrogen irradiation drives atomic reorganization, resulting in a granular coagulated texture. The progressive shift from sharp edges to blunt features observed in Figs. 6b–d indicates ion-induced microstructural changes. These irregular microstructure result from sharp temperature fluctuations caused by high-energy ion irradiation, leading to localized melting followed by rapid cooling [47].

At 24 shots (Fig. 6d), the film surface exhibits a more homogeneous texture with pronounced blunt edges and the near-complete disappearance of sharp facets. This transformation can be attributed to continuous carbon atom displacement and reattachment, where nitrogen ions act as active species inducing structural rearrangements. The high-energy ions dislodge carbon atoms from

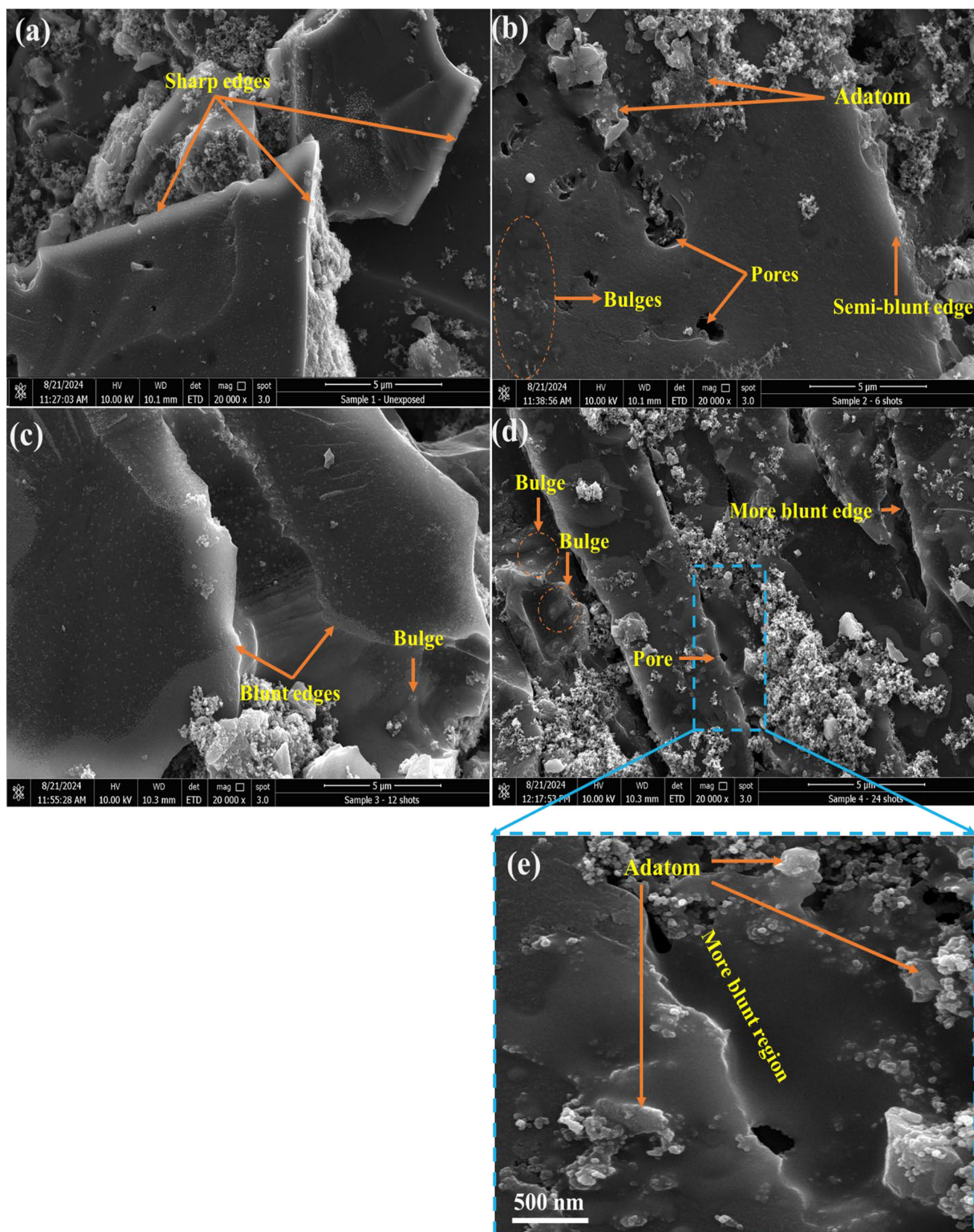


Fig. 6 FESEM micrographs of carbon thin films: **a** as-deposited sample, **b** after 6 nitrogen PF shots, **c** 12 shots, **d** 24 shots, and **e** enlarged view highlighting the blunter region in **(d)**

sharp edges, redistributing the material and leading to an overall blunt effect. Additionally, nitrogen atoms may be occupying interstitial positions or forming new chemical bonds with carbon [48], contributing to the structural modifications. The enlargement of the blunt region in Fig. 6e further emphasizes this effect, demonstrating the cumulative impact of nitrogen ion irradiation on surface morphology.

Another critical observation is the presence of dark bulges across the nitrogen-irradiated samples (Figs. 6b–d), which are absent in the as-deposited film (Fig. 6a). These bulges are likely caused by the high-energy impact of nitrogen ion beams, which induce localized heating on the carbon surface, leading to the emission of carbon particles that are subsequently redeposited in a random manner. Nitrogen ion bombardment (Fig. 6b and c) not only causes particle emission and the formation of surface bulges but also leads to the creation of adatoms. Adatoms are individual atoms displaced from their original lattice positions by the collision cascade but remain on the surface due to insufficient energy to escape or penetrate [49]. They contribute to surface roughness on an atomic scale. The presence of these bulges and adatoms suggests successful restructuring, reinforcing the idea that nitrogen ions play a crucial role in modifying the surface structure.

Nitrogen ion irradiation via the PF device transforms the surface from sharp-edged and rough to a compact, granular morphology with pores and bulges. This dual role of nitrogen ions in etching and restructuring enhances conductivity and electrochemical performance.

4.3 EDX analysis

EDX analysis was conducted to evaluate the elemental composition of the produced thin films, providing insights into nitrogen incorporation and overall film composition. Figures 7, 8, and 9, respectively, present the EDX spectra with atomic percentages, doping rate per shot and elemental mappings of the thin films subjected to different nitrogen irradiation shot counts, respectively. The presence of oxygen was detected across all samples, likely due to surface oxidation from ambient exposure or residual contamination from inadequate ultra-high vacuum conditions during analysis [50]. Notably, the nitrogen doping level is optimized at 7.93% in the sample exposed to 24 shots of nitrogen irradiation. Meanwhile, the 12-shot sample exhibits a nitrogen content of 5.96%, while the 6-shot sample achieves a doping level of 7.06%. These findings indicate nitrogen incorporation in the irradiated carbon films, supporting the potential of nitrogen ion irradiation as a technique for modifying the chemical characteristics of carbon-based thin films.

Interestingly, the nitrogen concentration was higher (7.06%) in the 6-shot sample than in the 12-shot sample (5.96%), suggesting a nonlinear relationship between shot count and nitrogen incorporation. This discrepancy is likely attributed to structural modifications induced by nitrogen ion irradiation exposure. Specifically, the increased porosity in the 6-shot sample (Fig. 6b) likely provided more active sites for nitrogen attachment. In contrast, the 12-shot sample exhibited lower nitrogen content, possibly due to surface etching effects [51, 52], that removed preexisting nitrogen species from the film. This suggests that nitrogen incorporation is influenced by a dynamic interplay between ion implantation, surface restructuring, and atomic displacement rather than following a simple shot-dependent trend. Additionally, the findings indicate that the nitrogen doping process in carbon thin films using a PF device occurs independently of the number of irradiation shots beyond a certain threshold.

Copper impurities were detected in all nitrogen-irradiated samples, with the highest concentration (2.11%) in the 24-shot sample (Fig. 7d). This increase is likely due to enhanced ablation of the PF anode, which intensifies with higher shot counts. High-energy ion beams generated during plasma pinch erode the anode surface, depositing its material onto the sample [22]. Such contamination is a known effect in PF systems.

Additionally, a notable fluorine peak was detected in all samples, originating from the PVDF binder used in the carbon slurry preparation. The highest fluorine content (12.06%) was observed in the as-deposited sample, while a gradual decrease was noted with increase in nitrogen irradiation, reaching a minimum of 4.94%. This reduction suggests that nitrogen ion irradiation effectively disrupts weaker C-F bonds, leading to fluorine reduction from the film surface. The interaction of high-energy nitrogen ions with the film likely promotes selective bond dissociation, particularly in regions where fluorine atoms are weakly bound. The decrease in fluorine content with increase in shot counts highlights the role of nitrogen irradiation in modifying the film's surface chemistry, enhancing its reactivity and suitability for functional applications.

Since nitrogen incorporation does not follow a strictly linear trend, it suggests that redistribution effects occur with increase in shot counts. Figure 8 shows the variation of nitrogen doping content per shot, ranging from 1.18% for 6 shots, 0.50% for 12 shots, and 0.33% for 24 shots. The doping content per shot is highest at lower shot counts, reaching 1.18% per shot for the 6-shot sample, and gradually decreases as more shots are applied. This indicates that nitrogen incorporation may saturate over time, leading to a reduced doping rate with increase in shot counts. As the surface becomes progressively enriched with nitrogen, the availability of active sites for further incorporation may diminish, resulting in a slower rate of nitrogen uptake at higher shot counts.

EDX elemental mapping of samples exposed to 6, 12, and 24 shots, as well as the as-deposited film, revealed significant morphological differences (Fig. 9). The elemental maps show a high concentration of carbon atoms in regions with large surface clusters (Figs. 9(ib, iib, iibb, and ivb)), indicating localized variations in deposition density. These surface morphological features may influence nitrogen incorporation, suggesting that nitrogen doping is affected not only by direct ion beam irradiation but also by restructuring effects at the surface. The nitrogen maps (Figs. 9(iid, iiic, and ivd)) exhibit a relatively uniform distribution across the sample interfaces, indicating that nitrogen atoms are effectively integrated into the carbon matrix rather than segregating into

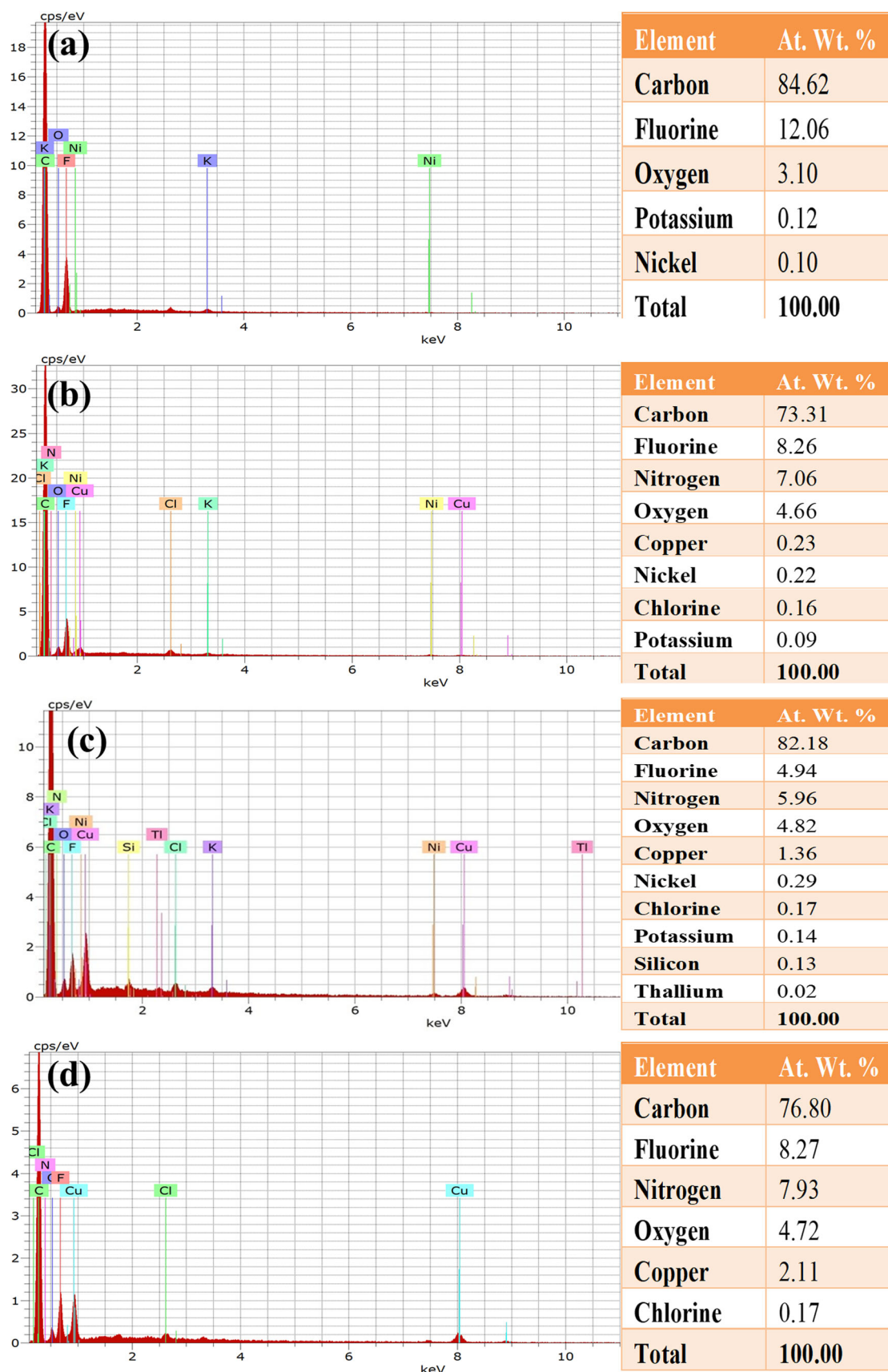


Fig. 7 EDX spectra illustrating elemental compositions for **a** as-deposited, **b** 6 shots, **c** 12 shots, and **d** 24 shots samples

Fig. 8 Variation of nitrogen doping content per shot (%) with increase in PF shot count (6, 12, and 24 shots) on carbon thin films

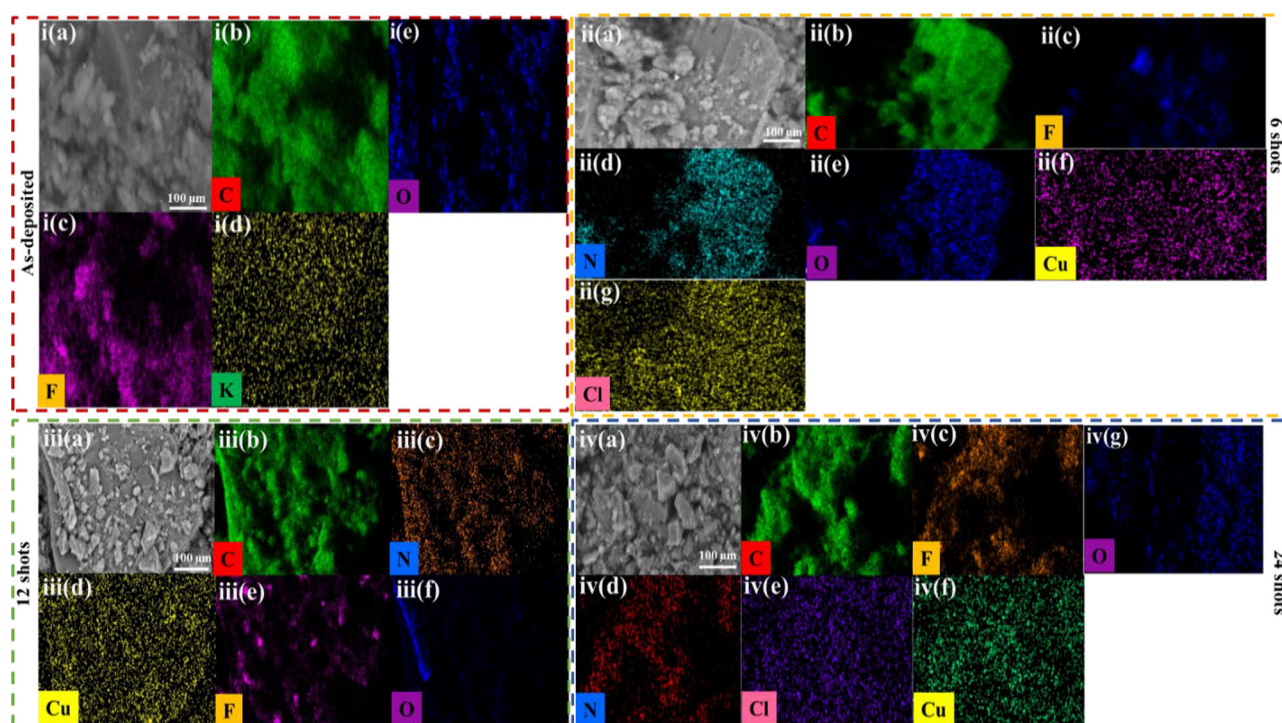
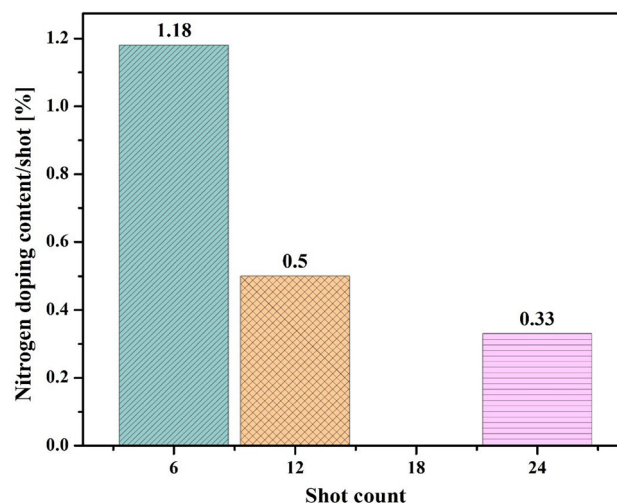


Fig. 9 EDX mapping of (i a–e) as-deposited, (ii a–g) 6 shots, (iii a–f) 12 shots, and (iv a–g) 24 shots samples

isolated domains. This uniformity contributes to the consistent chemical and physical properties of the films—an important factor for applications such as electrochemical energy storage where homogeneous doping is critical.

4.4 Functionalization analysis

FTIR analysis is utilized to identify the functional groups absorbed by carbon thin films as a result of nitrogen ion irradiation. Figure 10a presents the FTIR spectra within the 400 to 4000 cm^{-1} range, showcasing the diverse functional elements associated with the thin film. Figure 10b provides a magnified view of the highlighted region from Fig. 10a.

The FTIR spectrum shown in Fig. 10a features prominent peaks in all samples at 875, 1169, 1388, and 1526 cm^{-1} , ascribed to C-H bending, C-O stretching of esters, C-F stretching of a fluoro compound, and N-O stretching of a nitro compound [53], respectively. The pronounced fluoro-compound peak in all samples confirms the presence of the fluorinated compound PVDF, which was used as a binder during synthesis, a finding also supported by the EDX analysis (Fig. 7).

The peak at 2096 cm^{-1} in Fig. 9a is attributed to $\text{N}=\text{C}=\text{S}$ stretching of isothiocyanate. Peaks identified at 2149, 2155, 2165, and 2175 cm^{-1} in Fig. 10b are associated with $\text{S}-\text{C}\equiv\text{N}$ stretching of alkyne, while the peak at 2263 cm^{-1} represents $\text{C}\equiv\text{N}$ stretching

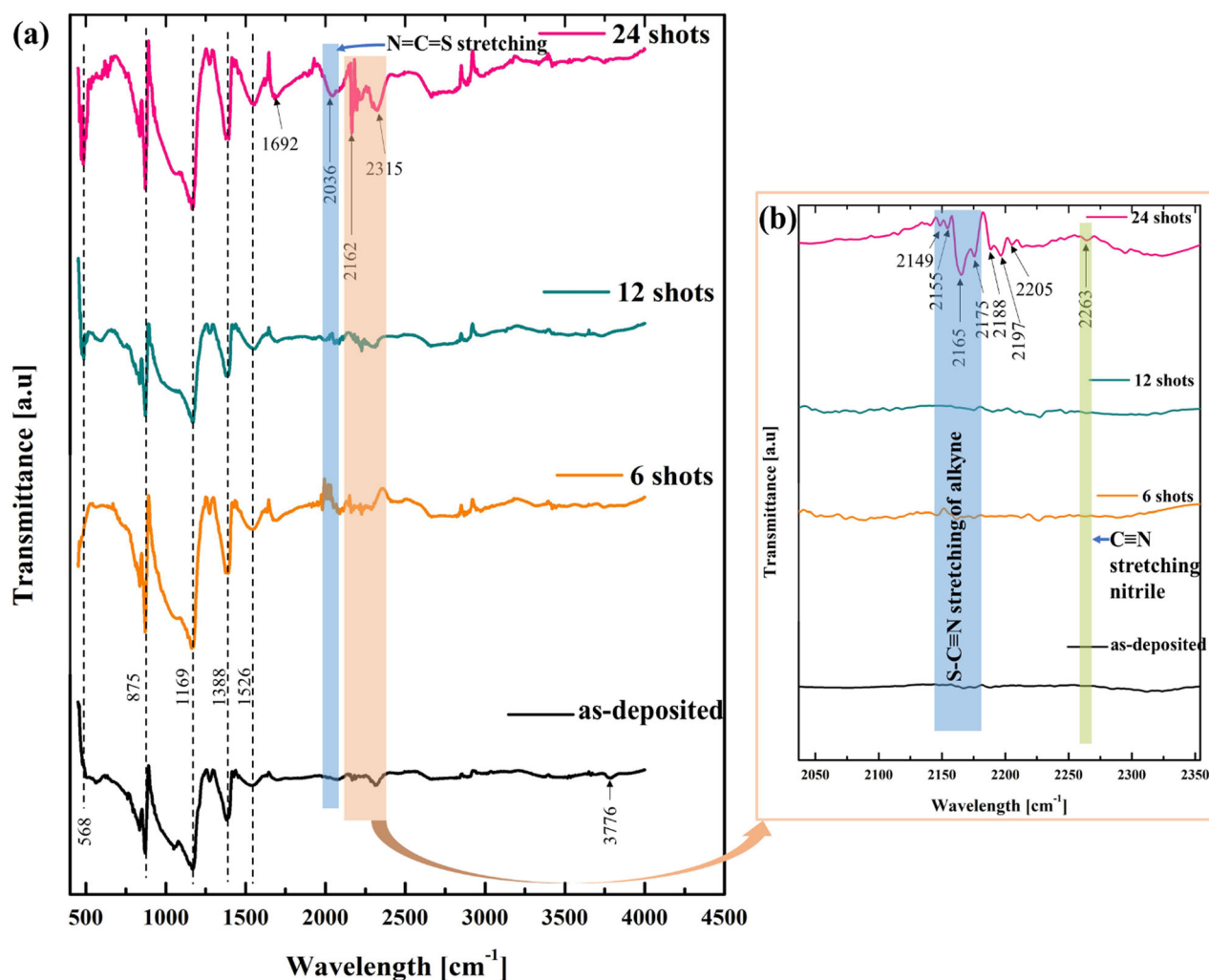


Fig. 10 **a** FTIR spectra of carbon thin films: as-deposited and after exposure to 6, 12, and 24 nitrogen ion shots, and **b** enlarged view of the selected spectral region from (a) for clearer comparison of functional group changes

of nitrile [52]. The formation of carbon–nitrogen bonds reduces sp^3 carbon content, disrupts carbon–carbon linkages, introduces voids, and contributes to the graphitization of the carbon films [54]. Additionally, energetic nitrogen atoms may disrupt the carbon rings, prompting carbon atoms to bond with sulfur contaminants. New carbon peaks at 2197 and 2205 cm^{-1} seen in Fig. 10b are associated with $C\equiv C$ stretching of alkyne.

As the number of shots on the thin film increases, the band at 2093 cm^{-1} (6 shots) shifts to lower frequencies of 2064 cm^{-1} (12 shots) and 2036 cm^{-1} (24 shots). These spectral shifts are likely due to the increasing defectiveness of the graphite-like structure as nitrogen atoms are incorporated into the carbon lattice, a similar effect observed by Ismagilov et al. [10]. The peak at 568 cm^{-1} , corresponding to C–Cl stretching of alkyl halides, becomes more pronounced as the number of shots increases, while the peak at 3776 cm^{-1} , associated with O–H stretching of alcohol, is gradually suppressed after 6 shots. Nitrogen doping may create reactive sites in the carbon structure, facilitating the incorporation of chlorine atoms and resulting in an increased concentration of C–Cl bonds, as evidenced by the Cl content in the EDX peak (Fig. 7). The suppression of the O–H peak is likely due to the gradual removal of oxygen-bonded impurities, as evidenced by the reduction in oxygen content in Fig. 7 after nitrogen irradiation.

The bonding of nitrogen to carbon atoms in the carbon thin film introduces defects in the structure [55], which are attached with multiple shots. This process results in a reduction in the intensity of the IR peak at 3776 cm^{-1} as the number of shots increases. The peaks completely disappear after 12 and 24 shots, indicating that multiple shots of nitrogen irradiation induce defects within the film, as confirmed by XRD analysis (Fig. 11). These defects become highly mobile, leading to the destruction of attachments along the film.

FTIR analysis identified nitrile, hydroxyl, and fluoro groups on the samples exposed to nitrogen irradiation, and these results were confirmed by EDX analysis (Fig. 7).

Fig. 11 XRD patterns of carbon thin films: comparison between the as-deposited sample and the sample irradiated with 24 nitrogen ion shots

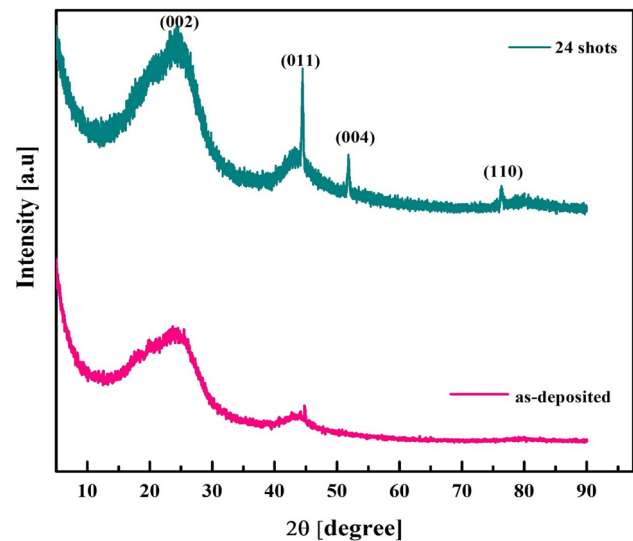


Table 2 Computed XRD parameters at (002) plane

Sample	FWHM, β (radians)	Particle size, D (nm)	Interlayer spacing, d_{002}
As-deposited	1.302	6.27	3.70 Å
24 shots	0.726	11.16	3.40 Å

4.5 Structural analysis

Figure 11 presents the XRD patterns for both the as-deposited and the 24-shot carbon samples, compared to the referenced patterns from JCPDS card no. 96–901–2231. The 24-shot sample was chosen for XRD analysis due to its optimal nitrogen doping content of 7.93%.

Pronounced peaks at ca. 25.0° and around 44.0° were observed for both samples, associated with the (002) and (011) inter-planes. Additionally, sharp peaks at 52° and 76°, associated with the (004) and (110) planes [56], respectively, appeared after 24 shots. The reflection at 25° (002) signifies the presence of small domains with aligned graphene sheets, while the reflections at 44° (011) may represent the plane of highly ordered carbon phases. These reflections are associated with fibrous or rounded structures composed of sp^2 hybridized carbon atoms [57, 58].

The peaks observed at around 52° and 76° in the 24-shot sample correspond to the (004) and (110) planes, respectively, which are indicative of graphite reflections. These peaks may be associated with new phases formed due to nitrogen incorporation, possibly including carbon nitrides or other nitrogen-induced compounds that emerge under nitrogen irradiation [59]. Nitrogen-doped graphene and other nitrogen-carbon hybrid materials often exhibit peak shifts or new peaks in this region due to the integration of nitrogen atoms into the carbon lattice [60]. A similar phenomenon was observed by Fajardo-Díaz et al. [61], who reported the emergence of new crystalline phases upon nitrogen doping, as evidenced by additional peaks in their XRD patterns.

In analyzing the structural differences between the as-deposited and 24-shot nitrogen ion-irradiated thin film samples, the (002) reflections were examined. Using Eq. (5), the crystallite sizes were computed as ~6.27 nm for the as-deposited sample and ~11.16 nm after 24 nitrogen ion shots, with corresponding full widths at half maximum (FWHM) of 1.302 and 0.726 radians (Table 2). The increase in crystallite size indicates enhanced structural ordering and grain growth likely due to nitrogen ion irradiation, which facilitates atomic mobility and reorganization of carbon nitride clusters.

The as-deposited sample exhibits an amorphous nature, characterized by broad XRD peaks, indicative of highly disordered carbon structures with minimal long-range order. In contrast, the 24-shot irradiated sample displays sharper and more intense diffraction peaks, suggesting enhanced crystallinity and the formation of more ordered carbon structures. The interaction between high-energy nitrogen ions and the carbon matrix induces atomic rearrangement [62]. The ion irradiation breaks weakly bonded structures within the carbon and facilitates the reorganization of carbon atoms into partial graphitic planes. This process is responsible for the observed increase in crystallinity in the 24-shot sample, as evidenced by increased crystallite size (~11.16 nm).

However, the estimated crystallite size increases from 6.27 nm to 11.16 nm, suggesting that nitrogen ion irradiation has promoted crystallite growth. This increase could be attributed to enhanced graphitization, defect reduction, or the coalescence of smaller crystalline domains at the nanoscale [63]. The crystallization of the carbon thin film with porous structures increases surface area, enhancing charge storage capacity by providing additional sites for ion adsorption and intercalation [64]. The presence of defects structures can facilitate ionic transport within the carbon matrix, improving overall ionic conductivity, which is crucial for fast charge and discharge rates.

The calculated interlayer spacing, d_{002} from Eq. (6), was determined to be 0.37 nm (3.70 Å) for the as-deposited sample and 0.340 nm (3.40 Å) for the 24-shot sample. The latter closely aligns with the typical graphite spacing of ~0.335 nm (3.35 Å) [65]. The reduction in interlayer spacing after 24 shots is attributed to nitrogen ion incorporation, which enhances atomic packing, induces strain, and modifies the lattice parameters, leading to interlayer contraction [66]. This suggests that the 24-shot carbon film remains partially graphitic [67], exhibiting misaligned layers rather than a fully ordered graphite structure. Our study indicates that disordered amorphous carbon can transition toward graphitization under PF-assisted nitrogen ion irradiation. This transformation involves a reduction in sp^3 bonding in favor of sp^2 hybridization at the high irradiation energy of 72.40 keV. Furthermore, our findings provide evidence of a noncoherent transformation from a stable amorphous phase to a metastable crystalline phase [68] under nonequilibrium conditions.

The slight expansion in spacing relative to fully crystalline graphite may result from structural defects, partial disorder, or weakened van der Waals interactions [69], indicating a transition toward graphitization. Notably, graphite with a slightly larger interplanar spacing facilitates faster Li-ion intercalation and deintercalation, improving high-rate discharge capability in Li-ion batteries [65]. These structural modifications may also enhance the material's electronic properties, making it promising for energy storage applications.

5 Conclusion

This study investigated the effects of nitrogen ion irradiation on carbon thin films synthesized using the doctor blade deposition method. The films were subjected to varying nitrogen irradiation shots (6, 12, and 24) using a 3.0 kJ dense PF device, operated at 2.54 kJ. The results demonstrated a significant alteration in the structural and elemental composition of the films, confirming successful nitrogen doping.

EDX analysis confirmed nitrogen doping, with a peak concentration of 7.93% at 24 shots, while FTIR spectroscopy validated the presence of nitrogen-related functional groups, including nitrile ($C\equiv N$) and isothiocyanate ($N=C=S$). Notably, the nitrogen content did not scale linearly with the number of irradiation shots, highlighting the intricate interplay between surface porosity and ion interaction dynamics in nitrogen retention. The 12-shot sample exhibited surface etching, contributing to a reduction in nitrogen content compared to the 6-shot sample. The nitrogen deposition rate was highest at lower shot counts (1.18% per shot for 6 shots) and gradually decreased with increase in shots (0.50% for 12 shots, 0.33% for 24 shots). This indicates that nitrogen incorporation reaches saturation over time, as the progressive occupation of active sites limits further nitrogen uptake at higher shot counts. Elemental mapping revealed a uniform nitrogen distribution, ensuring structural homogeneity—an essential factor for applications requiring controlled doping levels and material stability.

Beyond nitrogen incorporation, irradiation-induced modifications extended to impurity redistribution and bond disruption. The progressive decline in fluorine content with increase in nitrogen shots indicates the effective dissociation of C-F bonds from the PVDF binder, while copper impurities—originating from anode ablation—became more pronounced with higher irradiation doses.

XRD analysis provided further insight into the structural changes induced by nitrogen irradiation. The results indicated an increase in crystallite size with increase in nitrogen ion shots, with values rising from 6.27 nm in the as-deposited film to 11.16 nm after 24 shots. This suggests enhanced crystallinity due to surface restructuring induced by nitrogen ion beam interaction with the carbon surface. Notably, the sample irradiated with 24 shots exhibited additional diffraction peaks at 52° and 76° , corresponding to the (004) and (110) graphitic planes associated with nitrogen ion incorporation. The interlayer spacing, d_{002} , decreased from 0.37 nm (3.70 Å) in the as-deposited sample to 0.340 nm (3.40 Å) in the 24-shot sample due to nitrogen ion incorporation, aligning closely with the interlayer spacing of graphite. This study demonstrates that disordered amorphous carbon undergoes a transition toward graphitization when subjected to PF-assisted nitrogen ion irradiation. This structural modification suggests that nitrogen doping not only alters the elemental composition but also significantly enhances the crystallographic and graphitic properties of the films.

The findings of this study provide valuable insights into optimizing nitrogen doping and structural modifications of carbon thin films for potential applications in electronic devices and energy storage. Future work could investigate the effects of varying ion energies and plasma conditions to further enhance the tunability of these films. Additionally, incorporating fractal analysis or differential modeling approaches—such as surface diffusion models or ion-induced pattern formation equations—may provide a more rigorous description of the observed surface morphology dynamics. The insights gained from this research contribute to the advancement of PF-based doping and modification techniques, providing a rapid and efficient approach for enhancing carbon-based materials.

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Author contribution Haruna P. Wante was involved in conceptualization, methodology, research, formal analysis, and manuscript preparation. Seong Ling Yap helped in conceptualization, methodology, review, planning, and supervision. Kheng Y. Tsung contributed to data collection, editing, and analysis. Han Yi Tan was involved in editing and resource management. Lim Lian Kuang helped in planning and methodology. Sabastine C. Ezike contributed to review and formal analysis. Siaw Foon Lee helped in review and methodology.

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Data availability Data will be made available upon request.

Declarations

Conflict of interest The authors declare no personal or financial conflicts of interest affecting the research outcomes presented in this article.

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